

- ## **Topic:** Installation, Configuration, and Study of an Intrusion Detection System (IDS)

Objective

To install, configure, and analyze the functionality of Snort IDS for monitoring and detecting suspicious network activity.

Steps to Implement

1. What is Snort?

- Snort is a network intrusion detection and prevention system capable of performing packet logging, protocol analysis, and real-time traffic monitoring.
- It is an open-source IDS/IPS that monitors network traffic for malicious activity.

2. Installation of Snort

a. Update System

```
sudo apt update
sudo apt upgrade
```

b. Install Required Dependencies

```
sudo apt install -y snort
```

c. Verify Installation

```
dhondpratyay@lightning:~$ snort -V
,*> Snort! <*-
o"_)~ Version 2.9.15.1 GRE (Build 15125)
' ' By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
```

3. Configuration of Snort

a. Edit Configuration File

Open the Snort configuration file for editing:

```
sudo nano /etc/snort/snort.conf
```

Update:

~ Define your network's home IP range:

```
bash var HOME_NET 192.168.1.0/24
```

~ Include rule paths:

```
bash include $RULE_PATH/local.rules include $RULE_PATH/snort.rules
```

b. Create a Local Rules File

Define custom rules in:

```
sudo nano /etc/snort/rules/local.rules
```

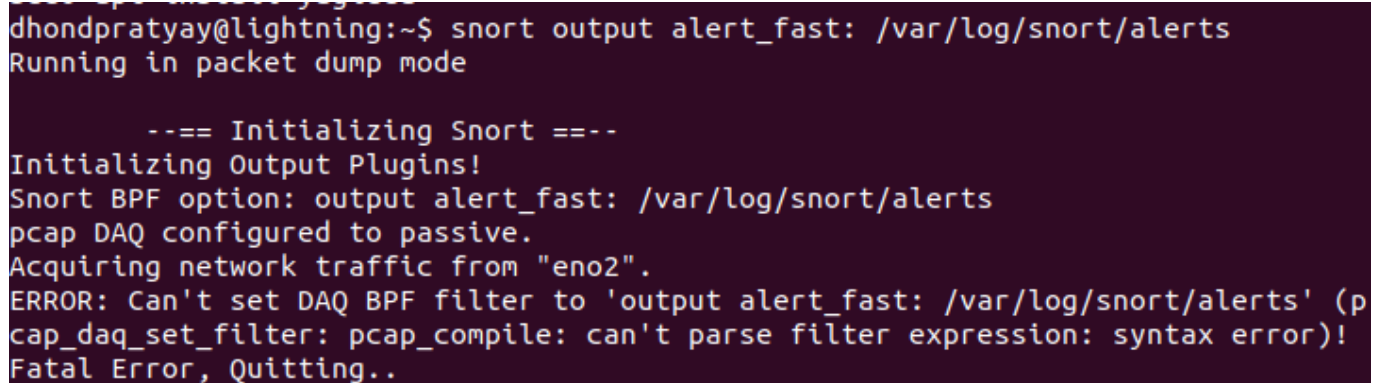
Example rule to alert on ICMP traffic:

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Traffic Detected";  
sid:1000001; rev:1;)
```

c. Enable Logging

Specify a logging directory in `/etc/snort/snort.conf`:

```
output alert_fast: /var/log/snort/alerts
```



```
dhondpratyay@lightning:~$ snort output alert_fast: /var/log/snort/alerts  
Running in packet dump mode  
  
--== Initializing Snort ==--  
Initializing Output Plugins!  
Snort BPF option: output alert_fast: /var/log/snort/alerts  
pcap DAQ configured to passive.  
Acquiring network traffic from "eno2".  
ERROR: Can't set DAQ BPF filter to 'output alert_fast: /var/log/snort/alerts' (p  
cap_daq_set_filter: pcap_compile: can't parse filter expression: syntax error)!  
Fatal Error, Quitting..
```

4. Running Snort

Run Snort in IDS mode:

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

(eth0 is an example; use your actual network interface name.)

```
dhondpratyay@lightning: ~  
dhondpratyay@lightning:~$ sudo snort -c /etc/snort/snort.conf -i eth0  
[sudo] password for dhondpratyay:  
Running in IDS mode  
  
--== Initializing Snort ==--  
Initializing Output Plugins!  
Initializing Preprocessors!  
Initializing Plug-ins!  
Parsing Rules file "/etc/snort/snort.conf"  
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 303  
7 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8  
088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 1137  
1 34443:34444 41080 50002 55555 ]  
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]  
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]  
PortVar 'SSH_PORTS' defined : [ 22 ]  
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]  
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]  
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301  
2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 802  
8 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9  
443 9999 11371 34443:34444 41080 50002 55555 ]  
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]  
Detection:  
  Search-Method = AC-Full-Q  
  Split Any/Any group = enabled  
  Search-Method-Optimizations = enabled  
  Maximum pattern length = 20  
Tagged Packet Limit: 256  
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsfe_engine.so... done  
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...  
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.  
  Finished Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules  
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_smtp_prepro  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_dns_preproc  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_dce2_prepro  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_dnp3_prepro  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_ssh_preproc  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_appid_prepr  
oc.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_ftptelnet_p  
reproc.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_sip_preproc  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_gtp_preproc  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_reputation_  
preproc.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_imap_prepro  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_sdf_preproc  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_pop_preproc  
c.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_modbus_prep  
roc.so... done  
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsfe_ssl_preproc  
c.so... done  
  Finished Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/  
Log directory = /var/log/snort  
WARNING: ip4 normalizations disabled because not inline.  
WARNING: tcp normalizations disabled because not inline.
```

```
WARNING: icmp4 normalizations disabled because not inline.
WARNING: ip6 normalizations disabled because not inline.
WARNING: icmp6 normalizations disabled because not inline.
Frag3 global config:
  Max frags: 65536
  Fragment memory cap: 4194304 bytes
Frag3 engine config:
  Bound Address: default
  Target-based policy: WINDOWS
  Fragment timeout: 180 seconds
  Fragment min_ttl: 1
  Fragment Anomalies: Alert
  Overlap Limit: 10
  Min fragment Length: 100
  Max Expected Streams: 768
Stream global config:
  Track TCP sessions: ACTIVE
  Max TCP sessions: 262144
```

```
dhondpratyay@lightning: ~
Max frags: 65536
Fragment memory cap: 4194304 bytes
Frag3 engine config:
  Bound Address: default
  Target-based policy: WINDOWS
  Fragment timeout: 180 seconds
  Fragment min_ttl: 1
  Fragment Anomalies: Alert
  Overlap Limit: 10
  Min fragment Length: 100
  Max Expected Streams: 768
Stream global config:
  Track TCP sessions: ACTIVE
  Max TCP sessions: 262144
  TCP cache pruning timeout: 30 seconds
  TCP cache nominal timeout: 180 seconds
  Memcap (for reassembly packet storage): 8388608
  Track UDP sessions: ACTIVE
  Max UDP sessions: 131072
  UDP cache pruning timeout: 30 seconds
  UDP cache nominal timeout: 180 seconds
  Track ICMP sessions: INACTIVE
  Track IP sessions: INACTIVE
  Log info if session memory consumption exceeds 1048576
  Send up to 2 active responses
  Wait at least 5 seconds between responses
  Protocol Aware Flushing: ACTIVE
  Maximum Flush Point: 16000
Stream TCP Policy config:
  Bound Address: default
  Reassembly Policy: WINDOWS
  Timeout: 180 seconds
  Limit on TCP Overlaps: 10
  Maximum number of bytes to queue per session: 1048576
  Maximum number of segs to queue per session: 2621
Options:
  Require 3-Way Handshake: YES
  3-Way Handshake Timeout: 180
  Detect Anomalies: YES
Reassembly Ports:
  21 client (Footprint)
  22 client (Footprint)
  23 client (Footprint)
  25 client (Footprint)
  42 client (Footprint)
  53 client (Footprint)
  79 client (Footprint)
  80 client (Footprint) server (Footprint)
  81 client (Footprint) server (Footprint)
  109 client (Footprint)
  110 client (Footprint)
  111 client (Footprint)
  113 client (Footprint)
  119 client (Footprint)
  135 client (Footprint)
  136 client (Footprint)
  137 client (Footprint)
  139 client (Footprint)
```

```

143 client (Footprint)
161 client (Footprint)
additional ports configured but not printed.
Stream UDP Policy config:
  Timeout: 180 seconds
HttpInspect Config:
  GLOBAL CONFIG
    Detect Proxy Usage: NO
    IIS Unicode Map Filename: /etc/snort/unicode.map
    IIS Unicode Map Codepage: 1252
    Memcap used for logging URI and Hostname: 150994944
    Max Gzip Memory: 104857600
    Max Gzip Sessions: 201649
    Gzip Compress Depth: 65535
    Gzip Decompress Depth: 65535
  DEFAULT SERVER CONFIG:
    Server profile: All
    Ports (PAF): 80 81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 434
3 4848 5250 6988 7000 7001 7144 7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8
123 8180 8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090 9091 9443 9999 11371 34443 34444 4
1080 50002 55555
    Server Flow Depth: 0
    Client Flow Depth: 0
    Max Chunk Length: 500000
    Small Chunk Length Evasion: chunk size <= 10, threshold >= 5 times
    Max Header Field Length: 750
    Max Number Header Fields: 100
    Max Number of WhiteSpaces allowed with header folding: 200

```

```

dhondpratyay@lightning: ~
Bad Message Direction Alert: DISABLED
Bad Payload Size Alert: DISABLED
Unrecognized Version Alert: DISABLED
Max Encrypted Packets: 20
Max Server Version String Length: 100
MaxClientBytes: 19600 (Default)
Ports:
  22
DCE/RPC 2 Preprocessor Configuration
Global Configuration
  DCE/RPC Defragmentation: Enabled
  Memcap: 102400 KB
  Events: co
  SMB Fingerprint policy: Disabled
Server Default Configuration
Policy: WinXP
Detect ports (PAF)
  SMB: 139 445
  TCP: 135
  UDP: 135
  RPC over HTTP server: 593
  RPC over HTTP proxy: None
Autodetect ports (PAF)
  SMB: None
  TCP: 1025-65535
  UDP: 1025-65535
  RPC over HTTP server: 1025-65535
  RPC over HTTP proxy: None
Invalid SMB shares: C$ D$ ADMIN$
Maximum SMB command chaining: 3 commands
SMB file inspection: Disabled
DNS config:
  DNS Client rdata txt Overflow Alert: ACTIVE
  Obsolete DNS RR Types Alert: INACTIVE
  Experimental DNS RR Types Alert: INACTIVE
  Ports: 53
SSLPP config:
  Encrypted packets: not inspected
  Ports:
    443      465      563      636      989
    992      993      994      995      7801
    7802      7900      7901      7902      7903
    7904      7905      7906      7907      7908
    7909      7910      7911      7912      7913
    7914      7915      7916      7917      7918
    7919      7920
  Server side data is trusted
  Maximum SSL Heartbeat length: 0
Sensitive Data preprocessor config:

```

```
dhondpratyay@lightning: ~  
ing old rule.  
4059 Snort rules read  
  3384 detection rules  
    0 decoder rules  
    0 preprocessor rules  
3384 Option Chains linked into 932 Chain Headers  
+++++  
-----[Rule Port Counts]-----  
|      tcp      udp      icmp      ip  
| src      151      18         0         0  
| dst     3306     126        0         0  
| any      383      48        54        22  
|  nc       27       8         17        20  
| s+d      12       5         0         0  
|-----  
-----[detection-filter-config]-----  
| memory-cap : 1048576 bytes  
-----[detection-filter-rules]-----  
| none  
|-----  
-----[rate-filter-config]-----  
| memory-cap : 1048576 bytes  
-----[rate-filter-rules]-----  
| none  
|-----  
-----[event-filter-config]-----  
| memory-cap : 1048576 bytes  
-----[event-filter-global]-----  
| none  
|-----  
-----[event-filter-local]-----  
| gen-id=1      sig-id=3152      type=Threshold tracking=src count=5      seconds=2  
| gen-id=1      sig-id=2924      type=Threshold tracking=dst count=10     seconds=60  
| gen-id=1      sig-id=2494      type=Both       tracking=dst count=20     seconds=60  
| gen-id=1      sig-id=2923      type=Threshold tracking=dst count=10     seconds=60
```

```

gen-id=1      sig-id=2523      type=Both      tracking=dst count=10  seconds=10
gen-id=1      sig-id=2496      type=Both      tracking=dst count=20  seconds=60
gen-id=1      sig-id=2495      type=Both      tracking=dst count=20  seconds=60
gen-id=1      sig-id=1991      type=Limit     tracking=src count=1    seconds=60
gen-id=1      sig-id=2275      type=Threshold tracking=dst count=5    seconds=60
gen-id=1      sig-id=3273      type=Threshold tracking=src count=5    seconds=2
-----[suppression]-----
| none
-----
Rule application order: pass->drop->sdrop->reject->alert->log
Verifying Preprocessor Configurations!
WARNING: flowbits key 'ms_sql_seen_dns' is checked but not ever set.
WARNING: flowbits key 'smb.tree.create.llsrpc' is set but not ever checked.
83 out of 1024 flowbits in use.

[ Port Based Pattern Matching Memory ]
+- [ Aho-Corasick Summary ] -----
Storage Format      : Full-Q
Finite Automaton   : DFA
Alphabet Size       : 256 Chars
Sizeof State        : Variable (1,2,4 bytes)
Instances           : 215
  1 byte states    : 204
  2 byte states    : 11
  4 byte states    : 0
Characters          : 64755
States              : 31951
Transitions         : 863868
State Density       : 10.6%
Patterns            : 5041
Match States        : 3836
Memory (MB)         : 16.90
  Patterns          : 0.51
  Match Lists       : 1.01
DFA
  1 byte states    : 1.02
  2 byte states    : 13.96
  4 byte states    : 0.00
-----
[ Number of patterns truncated to 20 bytes: 1038 ]
bcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x798a10839640 (11758)
ERROR: Can't start DAQ (-1) - SIOCGIFHWADDR: No such device!
Fatal Error, Quitting..
• dhondpratyay@lightning:~$

```

5. Study and Analysis

a. Generate Traffic

Test using **ping** or other tools to simulate malicious traffic:

```
ping 192.168.1.1
```

b. Review Alerts

Check the logs for alerts in:

```
cat /var/log/snort/alerts
```

Results

Snort successfully detected suspicious traffic based on predefined rules, demonstrating its capability to monitor and secure network environments.

Challenges and Solutions

Challenge: High false-positive rate due to overly generic rules.

Solution: Fine-tuned rules to target specific traffic patterns.

Conclusion

This exercise highlighted the importance of IDS systems like Snort in modern network security. With proper rule configuration, Snort acts as a reliable tool for detecting and preventing intrusions.

Future Work: Experiment with advanced features such as Snort's inline mode for intrusion prevention.

Feel free to adapt this report to match your observations during implementation!